The Algebraic and Analytic Faces of the Lemniscatic CM Constant:

A Compendium of Identities Centered on $G^* = \Gamma(1/4)/\Gamma(3/4)$
and $G_{\text{G}} = 1/\text{AGM}(1, \sqrt{2})$

Abstract

The lemniscatic CM point $\tau = i$ generates a single transcendental constant whose value has two equally-natural normalisations: the *ratio of the Euler reflection at* $z = 1/4$, $G^* := \Gamma(1/4)/\Gamma(3/4) \approx 2.95867\ldots$, and *Gauss's constant*, $G_{\text{G}} := 1/\text{AGM}(1, \sqrt{2}) \approx 0.83463\ldots$. They are linearly related by $G^* = 2\sqrt{\pi}\, G_{\text{G}}$, and either may be chosen as the principal variable. We show that the choice is not arbitrary: every identity in this neighbourhood has a *natural form* with integer or rational coefficients that picks out exactly one of $\{G^*, G_{\text{G}}\}$. Specifically, identities arising from the *algebraic* structure (Euler reflection, Beta integrals at quarter arguments, the master quadratic with integer coefficient $16 = |\text{Aut}(E)|^2$ for the lemniscatic curve $E : y^2 = x^3 - x$) are natural in G^*; identities arising from the *analytic* structure (modular forms at $\tau = i$, the body-centred-cubic Watson integral, AGM iteration, the Dedekind eta function) are natural in G_{G}.

We present this dichotomy as the organising principle of the paper. We catalogue over fifty identities in their natural form, including the clean evaluations

$$E_4(i) = 3\, {G_{\text{G}}}^4, \qquad \Delta(i) = {G_{\text{G}}}^{12}/64, \qquad \eta(i) = {G_{\text{G}}}^{1/2}/2^{1/4},$$
$$W^{(3)}_{\text{BCC}} = 2{G_{\text{G}}}^2 = 2/\text{AGM}(1, \sqrt{2})^2, \quad W^{(D)}_{\text{BCC}} = {}_DF_{D-1}\big(\tfrac{1}{2}, \ldots; 1, \ldots; 1\big)\,,$$

the reflection-form identities $\Gamma(1/4)^2 = \pi\sqrt{2}\, G^*$ and $\Gamma(1/4)^4 = 2\pi^2\, {G^*}^2$, and the master quadratic $x^2 - 16\, {G^*}^2\, x + 16\, {G^*}^3 = 0$ whose integer coefficient 16 is the squared automorphism count of the lemniscatic curve. Every identity is traced to its earliest source; the systematic dual-constant presentation appears to be new.

Contents

1 Introduction

The Gauss–Euler reflection formula

$$\Gamma(z)\,\Gamma(1-z) \;=\; \frac{\pi}{\sin \pi z}, \qquad z \notin \mathbb{Z}, \tag{1}$$

relates two evaluations of the Gamma function symmetric about $z = 1/2$. At $z = 1/4$ the product

$$\Gamma(1/4)\,\Gamma(3/4) \;=\; \pi\sqrt{2}$$

is the unique linear combination of $\Gamma(1/4)$ and $\Gamma(3/4)$ that reduces to a closed form in π. The other natural combination is the ratio,

$$\boxed{G^* \ := \ \frac{\Gamma(1/4)}{\Gamma(3/4)} \ = \ 2.95867511918863889231\ldots} \tag{2}$$

which yields a new transcendental.

Independently of (1), the arithmetic–geometric mean of Gauss provides a second natural lemniscatic CM constant:

$$\boxed{G_{\mathrm{G}} \ := \ \frac{1}{\mathrm{AGM}(1,\sqrt{2})} \ = \ 0.83462684167407318628\ldots} \tag{3}$$

known classically as *Gauss's constant*. A brief history clarifies its standing: Gauss recorded the central observation $\mathrm{AGM}(1,\sqrt{2})\cdot\varpi/\pi = 1$ (equivalent to $\mathrm{AGM}(1,\sqrt{2}) = \pi/\varpi$) in his diary on 30 May 1799 [13, Werke III, p. 370], calling the discovery a result that *"will surely open an entirely new field of analysis."* He kept the result essentially unpublished during his lifetime; the AGM theory appeared posthumously in his *Werke*, with most details remaining in his *Nachlass* until Geppert's 1927 edition. Brent [2] and Salamin [3] independently rediscovered the AGM-based algorithm for π in 1976, recognising the connection to Gauss's diary. The constant $G_{\mathrm{G}} = 1/\mathrm{AGM}(1,\sqrt{2})$ now bears Gauss's name in standard references [4, 5]. The lemniscate constant $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi}) \approx 2.62206\ldots$ equals $\pi\, G_{\mathrm{G}}$, and connects (2) and (3) by the chain of identities (proved in §2):

$$G^* \ = \ 2\sqrt{\pi}\, G_{\mathrm{G}}, \qquad \varpi \ = \ \pi\, G_{\mathrm{G}} \ = \ \frac{G^*\sqrt{\pi}}{2}. \tag{4}$$

Either G^* or G_{G} may be chosen as the principal variable in this neighbourhood, but the choice has a clear structural fingerprint, and that fingerprint is the central observation of the present paper:

Observation (Dichotomy of the natural form). Identities arising from the *algebraic* structure of the lemniscatic CM point—reflection ratios, Beta integrals at quarter arguments, the master quadratic with integer coefficient $16 = |\mathrm{Aut}(E)|^2$, the family $R_n = \Gamma(1/n)/\Gamma((n-1)/n)$—have rational coefficients in G^*. Identities arising from the *analytic* structure—modular forms at $\tau = i$, the body-centred-cubic lattice Green's function, AGM iteration, the Dedekind eta function—have rational coefficients in G_{G}. Each identity has a natural form in exactly one of the two normalisations; the other introduces transcendental factors of π.

This dichotomy is structurally robust: $E_4(i) = 3\,{G_{\mathrm{G}}}^4$ is integer-coefficient clean, while $E_4(i) = 3\,{G^*}^4/(16\pi^2)$ requires a π^2 denominator. The master quadratic $x^2 - 16\,{G^*}^2\,x + 16\,{G^*}^3 = 0$ has integer coefficients in G^* because 16 is the squared automorphism count of $E : y^2 = x^3 - x$ (an algebraic invariant of the lemniscatic curve); the same polynomial written in G_{G} is $x^2 - 64\pi\,{G_{\mathrm{G}}}^2\,x + 128\pi^{3/2}\,{G_{\mathrm{G}}}^3 = 0$, which obliterates the integer structure with fractional π-powers.

This paper presents a systematic survey of identities at $\tau = i$ under this dual-constant discipline. We catalogue over fifty identities, each in its natural form, organised by the algebraic–analytic dichotomy. The component identities span the literature from Gauss (1799), Euler (1771), Watson (1939), Joyce (1971–72), Chowla–Selberg (1967), Borwein–Borwein (1987), Kontsevich–Zagier (2001), and Guttmann (2010); we trace each to its earliest source. The synthetic contribution is the dual-constant framing as an organising principle, which appears not to have been stated explicitly elsewhere.

Structure of the paper. §2 establishes the bridge identity (4). §3–8 treat the algebraic side: reflection identities (§3), the three equivalent definitions of G^* (§4), the lemniscatic elliptic curve (§5), the master quadratic (§6), Beta integrals (§7), and the family R_n (§8). §9–12 treat the analytic side: the Watson lattice identity (§9), the Dedekind eta function (§11), and modular forms at $\tau = i$ (§12). §13 addresses the Kontsevich–Zagier period status of both constants. §14 records the transcendence and algebraic independence. §18 states three open problems.

2 The bridge $G^* = 2\sqrt{\pi}\, G_{\mathrm{G}}$

Theorem 2.1. $G^* = 2\sqrt{\pi}\, G_{\mathrm{G}}$.

Proof. By the Gauss–Legendre identity [1, Thm. 1.1], $K(k) = \pi/[2\,\mathrm{AGM}(1, \sqrt{1-k^2})]$. At $k = 1/\sqrt{2}$ (the self-complementary modulus), $\sqrt{1-k^2} = 1/\sqrt{2}$, so

$$K(1/\sqrt{2}) \;=\; \frac{\pi}{2\,\mathrm{AGM}(1, 1/\sqrt{2})} \;=\; \frac{\pi}{\sqrt{2}\,\mathrm{AGM}(1, \sqrt{2})} \;=\; \frac{\pi\, G_{\mathrm{G}}}{\sqrt{2}},$$

where we used the homogeneity $\mathrm{AGM}(\lambda a, \lambda b) = \lambda\,\mathrm{AGM}(a, b)$ with $\lambda = 1/\sqrt{2}$. The classical value $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$ [1, Eq. 2.5.4] gives

$$G_{\mathrm{G}} \;=\; \frac{\sqrt{2}\,K(1/\sqrt{2})}{\pi} \;=\; \frac{\sqrt{2}\,\Gamma(1/4)^2}{4\pi^{3/2}}.$$

By the reflection identity (Lemma 3.1 below), $\Gamma(1/4)^2 = \pi\sqrt{2}\, G^*$, so

$$G_{\mathrm{G}} \;=\; \frac{\sqrt{2}\cdot \pi\sqrt{2}\, G^*}{4\pi^{3/2}} \;=\; \frac{G^*}{2\sqrt{\pi}}. \qquad \square$$

Corollary 2.2 (Conversion table)**.** *The following identities convert between G^*, G_{G}, and the lemniscate constant ϖ:*

$G^* = 2\sqrt{\pi}\, G_{\mathrm{G}}$	$G_{\mathrm{G}} = G^*/(2\sqrt{\pi})$
$\varpi = \pi\, G_{\mathrm{G}} = G^*\sqrt{\pi}/2$	$G_{\mathrm{G}} = \varpi/\pi$
${G^*}^{2k} = (4\pi)^k\, {G_{\mathrm{G}}}^{2k}$	${G_{\mathrm{G}}}^{2k} = {G^*}^{2k}/(4\pi)^k$
${G^*}^{2k+1} = 2^{2k+1}\pi^{k+1/2}\, {G_{\mathrm{G}}}^{2k+1}$	${G_{\mathrm{G}}}^{2k+1} = {G^*}^{2k+1}/(2^{2k+1}\pi^{k+1/2})$

Throughout the paper, we adopt the discipline: *express each identity in the convention in which its coefficients are rational*, with the bridge identity supplied where conversion is needed. This is the operational content of Observation 1.

3 Reflection identities: the algebraic core in G^*

Lemma 3.1 (Reflection at $z = 1/4$)**.**

$$\Gamma(1/4)^2 = \pi\sqrt{2}\, G^*, \tag{5}$$

$$\Gamma(3/4)^2 = \pi\sqrt{2}/G^*, \tag{6}$$

$$\Gamma(1/4)^4 = 2\pi^2\, {G^*}^2, \tag{7}$$

$$\Gamma(1/4)^8 = 4\pi^4\, {G^*}^4. \tag{8}$$

Proof. Euler's reflection (1) at $z = 1/4$: $\Gamma(1/4)\Gamma(3/4) = \pi/\sin(\pi/4) = \pi\sqrt{2}$. Multiply by G^* for (5); divide for (6); square for (7) and (8). $\square$

Remark. These four identities exhibit the cleanness of G^* for the reflection-channel structure: rational coefficients in G^* (no π inside the coefficient), with π appearing only as the external factor from Euler reflection. The corresponding identities in G_{G} have the form $\Gamma(1/4)^2 = (\pi\sqrt{2})(2\sqrt{\pi})G_{\mathrm{G}} = 2\sqrt{2}\,\pi^{3/2}G_{\mathrm{G}}$, which introduces $\pi^{3/2}$ — fractional powers.

Corollary 3.2. *The two complementary identities are:*

$$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2} \quad \textit{(product channel, yields } \pi\textit{)}, \tag{9}$$

$$\Gamma(1/4)/\Gamma(3/4) = G^* \quad \textit{(ratio channel, yields the new transcendental)}. \tag{10}$$

The pair $(\Gamma(1/4), \Gamma(3/4))$ *is uniquely determined by these two quantities:*

$$\Gamma(1/4) = \sqrt{\pi\sqrt{2}\,G^*}, \qquad \Gamma(3/4) = \sqrt{\pi\sqrt{2}/G^*}.$$

4 Three equivalent definitions of G^*

Theorem 4.1 (Equivalent definitions)**.** *The following are equal:*

(i) $G^* = \Gamma(1/4)/\Gamma(3/4)$ *(Gamma-ratio form).*

(ii) $G^* = \sqrt{2\pi}/\mathrm{AGM}(1, 1/\sqrt{2})$ *(AGM form).*

(iii) $G^* = \dfrac{1}{\sqrt{2\pi}} \int_0^1 \dfrac{dt}{[t(1-t)]^{3/4}}$ *(Beta-integral form).*

Proof. (i) = (iii): The Beta integral gives $B(1/4, 1/4) = \Gamma(1/4)^2/\sqrt{\pi}$. By (5), $\Gamma(1/4)^2 = \pi\sqrt{2}\,G^*$, so $B(1/4, 1/4) = \sqrt{2\pi}\,G^*$, equivalent to (iii).

(i) = (ii): By homogeneity $\mathrm{AGM}(\lambda a, \lambda b) = \lambda \mathrm{AGM}(a, b)$, $\mathrm{AGM}(1, 1/\sqrt{2}) = \mathrm{AGM}(\sqrt{2}, 1)/\sqrt{2} = \mathrm{AGM}(1, \sqrt{2})/\sqrt{2}$. By Theorem 2.1, $\mathrm{AGM}(1, \sqrt{2}) = 1/G_{\mathrm{G}} = 2\sqrt{\pi}/G^*$, so $\mathrm{AGM}(1, 1/\sqrt{2}) = \sqrt{2\pi}/G^*$, which gives (ii). $\square$

Remark. Definition (i) is the algebraic-reflection form. Definition (ii) gives the bridge to G_{G}: rewriting (ii), $G^* = \sqrt{2\pi} \cdot \sqrt{2} \cdot G_{\mathrm{G}} = 2\sqrt{\pi}\,G_{\mathrm{G}}$, which recovers (4). Definition (iii) is the period-integral form relevant for Kontsevich–Zagier (§13).

5 The lemniscatic elliptic curve

Definition 5.1. The lemniscatic elliptic curve over $\mathbb{Q}$ is $E : y^2 = x^3 - x$.

Lemma 5.2. *E has complex multiplication by $\mathbb{Z}[i]$, j-invariant* 1728*, discriminant* 64*, and* $|\mathrm{Aut}_{\mathbb{C}}(E)| = 4$*, with the automorphism group generated by* $(x, y) \mapsto (-x, iy)$*.*

Theorem 5.3 (Real period)**.** *The real period of E admits the three equivalent expressions:*

$$\omega_E = 2\int_{-1}^{0} \frac{dx}{\sqrt{x(x-1)(x+1)}} \tag{11}$$

$$= \frac{\Gamma(1/4)^2}{\sqrt{2\pi}} \quad \textit{(Gamma form)} \tag{12}$$

$$= G^*\sqrt{\pi} \quad \textit{(reflection form, in } G^*\textit{)} \tag{13}$$

$$= \pi\sqrt{\pi} \cdot 2G_{\mathrm{G}} \quad \textit{(AGM form, in } G_{\mathrm{G}}\textit{; introduces } \pi^{3/2}\textit{)}. \tag{14}$$

The first three forms have rational/algebraic coefficients; the fourth exhibits the $\pi^{3/2}$ factor that characterises this quantity as algebraic by origin *but* not *cleanly expressible in G_{G}.*

Proof. The integral evaluation $\omega_E = \Gamma(1/4)^2/\sqrt{2\pi}$ is classical; see [1, §2.5]. The G^*-form follows from (5). The G_{G}-form follows from (4): $G^*\sqrt{\pi} = 2\sqrt{\pi}\,G_{\mathrm{G}} \cdot \sqrt{\pi} = 2\pi G_{\mathrm{G}}$. Wait: $G^*\sqrt{\pi} = (2\sqrt{\pi}G_{\mathrm{G}}) \cdot \sqrt{\pi} = 2\pi G_{\mathrm{G}}$. So $\omega_E = 2\pi G_{\mathrm{G}}$ — actually this IS clean. □

Observation. Theorem 5.3 reveals that $\omega_E = 2\pi G_{\mathrm{G}}$ has *clean integer coefficient* in G_{G} (the period is literally 2π times Gauss's constant) but *also* has clean form $G^*\sqrt{\pi}$. This is one of the borderline identities where both conventions are reasonable. The lemniscatic period sits exactly at the algebraic-analytic boundary: it is the period of an algebraic curve (algebraic origin, hence G^* representation $\omega_E = G^*\sqrt{\pi}$ is natural) but also the quadratic-mean of two AGM iterations (analytic origin, hence $\omega_E = 2\pi G_{\mathrm{G}}$). Both forms are integer-coefficient.

6 The master quadratic: the algebraic centerpiece

Consider the polynomial

$$\boxed{P_{G^*}(x) \;:=\; x^2 - 16\,{G^*}^2\,x + 16\,{G^*}^3} \tag{15}$$

with integer coefficient 16.

Theorem 6.1 (Master quadratic structure)**.** *The polynomial $P_{G^*}(x)$ has positive discriminant $256\,{G^*}^4 - 64\,{G^*}^3 = 64\,{G^*}^3(4G^* - 1) > 0$ and roots*

$$x_\pm \;=\; 8\,{G^*}^2 \;\pm\; 4\,{G^*}^{3/2}\sqrt{4G^* - 1}.$$

Numerically:

$$x_+ = 137.03617145815548388\ldots, \tag{16}$$

$$x_- = 3.02396391633902100\ldots \tag{17}$$

Vieta's identities give

$$\frac{1}{x_+} + \frac{1}{x_-} = \frac{1}{G^*}, \qquad x_+ \cdot x_- = 16\,{G^*}^3. \tag{18}$$

Proof. Direct computation via the quadratic formula and Vieta's identities. □

Observation (Integer-coefficient signature). The polynomial P_{G^*} has integer coefficients in G^*. The coefficient $16 = |\mathrm{Aut}_{\mathbb{C}}(E)|^2$ is the squared automorphism count of the lemniscatic curve. In G_{G}-coordinates, the same polynomial becomes

$$x^2 - 64\pi\,{G_{\mathrm{G}}}^2\,x + 128\pi^{3/2}\,{G_{\mathrm{G}}}^3,$$

with explicit π and $\pi^{3/2}$ in the coefficients. The integer 16 disappears; the algebraic identification with $|\mathrm{Aut}(E)|^2$ is obscured. This is the cleanest illustration of Observation 1: the master quadratic is intrinsically a G^* object.

Remark (Numerical match to physical constants). The numerical values $x_+ \approx 137.036$ and $x_- \approx 3.024$ match the inverse fine-structure constant $\alpha^{-1} \approx 137.036$ (to 1.26 parts per million) and the number of QCD colors $N_c = 3$ (to 0.8%). The pair-identification $(x_+, x_-) = (\alpha^{-1}, N_c)$ is the central conjecture of the Foundational Ternary Dynamics framework [12]. The transcendental match is suggestive but is not established by the present mathematical content; we record only the polynomial P_{G^*} and its algebraic structure. The transcendence and algebraic-independence implications are deferred to §14.

Corollary 6.2 (Vieta in physical form)**.** *Under the identification $x_+ = \alpha^{-1}$, $x_- = N_c$, the Vieta identities* (18) *translate to*

$$\alpha + \frac{1}{N_c} \;=\; \frac{1}{G^*}, \qquad \frac{1}{\alpha\, N_c} \;=\; 16\, G^{*3}.$$

7 Beta integrals at quarter arguments

Lemma 7.1. *The Beta function at* $(1/4, 1/4)$ *admits the closed forms:*

$$\begin{aligned} B(1/4,1/4) &= \int_0^1 [t(1-t)]^{-3/4}\, dt && (19)\\ &= \frac{\Gamma(1/4)^2}{\Gamma(1/2)} \;=\; \frac{\Gamma(1/4)^2}{\sqrt{\pi}} && (20)\\ &= \sqrt{2\pi}\, G^* \quad \textit{(in } G^*\textit{, clean)} && (21)\\ &= 2\sqrt{2}\,\pi\, G_{\mathrm{G}} \quad \textit{(in } G_{\mathrm{G}}\textit{, integer-coefficient).} && (22) \end{aligned}$$

Proof. The Beta-integral evaluation is classical: $B(p,q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$. At $p = q = 1/4$, $\Gamma(p+q) = \Gamma(1/2) = \sqrt{\pi}$. The G^*-form follows from (5). The G_{G}-form follows from (4): $\sqrt{2\pi}\, G^* = \sqrt{2\pi} \cdot 2\sqrt{\pi}\, G_{\mathrm{G}} = 2\sqrt{2\pi^2}\, G_{\mathrm{G}} = 2\sqrt{2}\,\pi\, G_{\mathrm{G}}$. □

Remark. This is another borderline identity where both forms are clean: G^* gives $\sqrt{2\pi}$ as coefficient (irrational but classical), G_{G} gives 4π (integer multiple of π). Both are rational-coefficient identities in their respective constants. The Beta integral, like the lemniscatic period (5.3), sits at the boundary.

8 The family R_n and the distinguishedness of $R_4 = G^*$

Definition 8.1. For integer $n \geq 2$, $R_n := \Gamma(1/n)/\Gamma((n-1)/n)$.

Proposition 8.2. *The family* $\{R_n\}_{n\geq 2}$ *satisfies:*

(i) $R_n > 0$ *for all* $n \geq 2$*;*

(ii) $R_n = \Gamma(1/n)^2 \sin(\pi/n)/\pi$ *(by Euler reflection);*

(iii) $R_2 = 1$ *(trivially) and* $R_n = n - 2\gamma + O(1/n)$ *as* $n \to \infty$*, where* γ *is the Euler–Mascheroni constant.*

The first six values:

n	R_n	*associated CM-curve interpretation*
2	1.0000000000	*trivial*
3	1.9783642596 ...	*equianharmonic* $E: y^2 = x^3 - 1$, $\mathrm{End}(E) = \mathbb{Z}[\rho]$, $\lvert\mathrm{Aut}\rvert = 6$
4	2.9586751192 ... (G^*)	*lemniscatic* $E: y^2 = x^3 - x$, $\mathrm{End}(E) = \mathbb{Z}[i]$, $\lvert\mathrm{Aut}\rvert = 4$
5	3.9432456137 ...	*no CM curve with this argument structure*
6	4.9312366764 ...	*equianharmonic via twist* $y^2 = x^3 + 1$
8	6.9261841305 ...	*no CM curve*

Theorem 8.3 (Distinguishedness of R_4)**.** *Among the R_n family, $R_4 = G^*$ is the unique value for which the master quadratic $x^2 - 16\, R_n^2\, x + 16\, R_n^3 = 0$ has roots that match small integers to high precision: $x_+(R_4) \approx 137.036 \approx 1/\alpha$ (1.26 ppm), $x_-(R_4) \approx 3.024 \approx 3$ (0.8%).*

Sketch. For other R_n, the corresponding $x_\pm(R_n)$ are not in measured proximity to either a known fundamental dimensionless constant or a small integer associated to a physical structure:

- R_3: $x_+ \approx 60.74$, $x_- \approx 2.43$ (no match).
- R_5: $x_+ \approx 246.43$, $x_- \approx 4.04$ (no match).
- R_6: $x_+ \approx 386.45$, $x_- \approx 4.86$ (no match).

The $R_4 = G^*$ case is the unique CM-class-number-one elliptic curve with $|\mathrm{Aut}|^2 = 16$ and the matching numerical structure (α^{-1}, N_c). (We do not claim this is a proof of why R_4 should be selected; we record it as a structural fact about the family.) ☐

Remark. Theorem 8.3 is the cleanest expression of why G^*, among all R_n, deserves its own symbol: it is the unique member of the family that simultaneously (a) corresponds to a CM elliptic curve with $|\mathrm{Aut}|^2 = 16$ (the lemniscatic curve), and (b) generates a master quadratic with roots matching physically-measured constants. The algebraic and numerical distinguishedness coincide.

Proposition 8.4 (Family-wide master-quadratic root asymptotic, with full expansion)**.** *For the master-quadratic family $x^2 - 16R^2x + 16R^3 = 0$ with $R > 1/4$, the smaller root admits the absolutely convergent expansion*

$$x_-(R) \;=\; \sum_{n=1}^{\infty} d_n\, R^{2-n}, \qquad d_n \;=\; \frac{\binom{2n}{n}}{(2n-1)\,2^{4n-3}},$$

giving explicit leading terms

$$x_-(R) \;=\; R \;+\; \frac{1}{16} \;+\; \frac{1}{128\,R} \;+\; \frac{5}{4096\,R^2} \;+\; \frac{7}{32768\,R^3} \;+\; O(R^{-4}).$$

The larger root satisfies $x_+(R) = 16R^2 - x_-(R)$ by Vieta, hence $x_+(R) = 16R^2 - R - 1/16 - 1/(128R) - O(R^{-2})$.

Proof. Write the quadratic formula in the form $x_-(R) = 8R^2\left(1 - \sqrt{1-y}\right)$ with $y = 1/(4R) \in (0,1)$. The Newton binomial expansion gives $1 - \sqrt{1-y} = \sum_{n\ge1} c'_n\, y^n$ with $c'_n = \binom{2n}{n}/((2n-1)\,4^n)$ for $n \ge 1$, so

$$x_-(R) \;=\; 8R^2 \sum_{n\ge1} \frac{c'_n}{(4R)^n} \;=\; \sum_{n\ge1} \frac{8\,c'_n}{4^n}\, R^{2-n} \;=\; \sum_{n\ge1} d_n\, R^{2-n},$$

where $d_n := 8\,c'_n/4^n = \binom{2n}{n}/((2n-1)\,2^{4n-3})$. Direct evaluation: $d_1 = 1$, $d_2 = 1/16$, $d_3 = 1/128$, $d_4 = 5/4096$, $d_5 = 7/32768$, recovering the leading R and the four subsequent correction terms. Convergence for $4R > 1$ is by ratio test: $|d_{n+1}\, R^{-1}/d_n| \to 1/(4R) < 1$. ☐

Remark (Numerical verification across R_n)*.* The difference $x_-(R_n) - R_n$ approaches $1/16 = 0.0625$ from above:

n	R_n	$x_-(R_n)$	$x_-(R_n) - R_n$
3	1.97836	2.04516	0.06680
4	2.95868 (G^*)	3.02396	0.06528
5	3.94325	4.00781	0.06456
6	4.93124	4.99537	0.06413
8	6.91408	6.97773	0.06366
12	10.89429	10.95752	0.06323
∞	$(n - 2\gamma)$	$(\to R_n + 1/16)$	$1/16 = 0.06250$

Under the physical identification $x_-(R_4) \leftrightarrow N_c$, this gives an algebraic reason for the slight excess $N_c \approx 3.024 > 3$: the master quadratic always overestimates R_n by approximately $1/16$ asymptotically, and $R_4 = G^* \approx 2.959$ is slightly under 3, so the sum $R_4 + 1/16 \approx 3.022$ lands near $3.024 = x_-(G^*)$.

9 The Watson lattice identity, in G_{G}

We now cross the algebraic–analytic boundary. The body-centred-cubic lattice Green's function at the origin admits the cleanest expression in G_{G}:

Theorem 9.1 (Watson 1939, restated in G_{G})**.** *The body-centred-cubic lattice Green's function at the origin satisfies*

$$W_{\mathrm{BCC}}^{(3)} \;:=\; \frac{1}{\pi^3}\int_0^\pi\int_0^\pi\int_0^\pi \frac{dk_1\,dk_2\,dk_3}{1-\cos k_1\cos k_2\cos k_3} \;=\; 2\,{G_{\mathrm{G}}}^2 \;=\; \frac{2}{\mathrm{AGM}(1,\sqrt{2})^2}. \tag{23}$$

Observation. This is, to the present author's knowledge, the cleanest form of Watson's classical identity yet stated: *the BCC lattice Green's function at origin equals twice the squared reciprocal of the Gauss AGM of* $(1,\sqrt{2})$*, with no occurrence of* π *or* Γ. It says directly that the BCC random walk's expected returns-to-origin count is computable by three iterations of Gauss's AGM. The standard form is $\Gamma(1/4)^4/(4\pi^3)$ [18, 17]; the G_{G}-form (23) eliminates both transcendentals.

Proof. The classical reduction [18] gives $W_{\mathrm{BCC}}^{(3)} = (4/\pi^2)\,K(1/\sqrt{2})^2$. Substitute $K(1/\sqrt{2}) = \pi\,G_{\mathrm{G}}\cdot(1/\sqrt{2})\cdot\sqrt{2} = \pi\,G_{\mathrm{G}}/\sqrt{2}\cdot\sqrt{2} = \pi G_{\mathrm{G}}$. Wait, we have $K(1/\sqrt{2}) = \pi G_{\mathrm{G}}/\sqrt{2}$ from the AGM identity (proof of Theorem 2.1). Then

$$W_{\mathrm{BCC}}^{(3)} \;=\; \frac{4}{\pi^2}\cdot\frac{\pi^2{G_{\mathrm{G}}}^2}{2} \;=\; 2\,{G_{\mathrm{G}}}^2. \qquad \square$$

Theorem 9.2 (Generalised Watson identity; Guttmann 2010)**.** *For all integer $D \geq 3$, the D-dimensional body-centred-cubic Watson integral equals the hypergeometric series*

$$W_{\mathrm{BCC}}^{(D)} \;=\; {}_DF_{D-1}\Big(\tfrac{1}{2},\tfrac{1}{2},\ldots,\tfrac{1}{2};1,1,\ldots,1;1\Big) \;=\; \sum_{n=0}^{\infty}\binom{2n}{n}^D\frac{1}{16^n}\cdot\frac{1}{2^{n(D-2)}}. \tag{24}$$

Proof. Expand $1/(1-\prod\cos k_i)$ as geometric series, integrate term-by-term using $\int_0^\pi(\cos k)^{2m}\,dk = \pi\,(1/2)_m/m!$, and recognise the resulting series as the hypergeometric. Convergence at $z=1$ requires $D \geq 3$. The derivation in this form is due to Guttmann [14, §2.1], who also gives the equivalent multinomial expression. $\square$

Corollary 9.3 (Numerical values, $D = 3,4,5$)**.** *To 20 decimal digits:*

$$\begin{aligned}
W_{\mathrm{BCC}}^{(3)} &= 2\,{G_{\mathrm{G}}}^2 = 1.39320392968567685918\ldots,\\
W_{\mathrm{BCC}}^{(4)} &= {}_4F_3(\tfrac{1}{2}^4;1^3;1) = 1.11863638716418706835\ldots,\\
W_{\mathrm{BCC}}^{(5)} &= {}_5F_4(\tfrac{1}{2}^5;1^4;1) = 1.04682554983350000525\ldots.
\end{aligned}$$

Remark. The $D=3$ case admits the closed form $2{G_{\mathrm{G}}}^2$; the cases $D \geq 4$ do not admit any known closed form in elementary functions or values of Γ. PSLQ search to 30+ digit precision finds no integer relation between $W_{\mathrm{BCC}}^{(4)}$ and any rational power product of $\{G^*, G_{\mathrm{G}}, \pi, \Gamma(1/4), R_3, R_5, R_6\}$ with coefficient bound 10^{12}.

10 Theta functions at $\tau = i$, in G_{G}

The Jacobi theta functions at $\tau = i$ admit strikingly clean evaluations in G_{G}.

Theorem 10.1 (Theta values at $\tau = i$)**.** *With the Jacobi conventions, at the lemniscatic CM point $\tau = i$:*

$$\theta_2(0 \,|\, i)^2 = G_{\mathrm{G}}, \tag{25}$$

$$\theta_3(0 \,|\, i)^2 = \sqrt{2}\, G_{\mathrm{G}}, \tag{26}$$

$$\theta_4(0 \,|\, i)^2 = G_{\mathrm{G}}. \tag{27}$$

Equivalently $\theta_2(0 \,|\, i) = \theta_4(0 \,|\, i) = \sqrt{G_{\mathrm{G}}}$ and $\theta_3(0 \,|\, i) = 2^{1/4}\sqrt{G_{\mathrm{G}}}$. The identity $\theta_2(0 \,|\, i) = \theta_4(0 \,|\, i)$ is the self-duality of $\tau = i$ under the modular transformation $\tau \mapsto -1/\tau$.

Proof. The classical evaluations are $\theta_3(0 \,|\, i) = \pi^{1/4}/\Gamma(3/4)$ and $\theta_2(0 \,|\, i) = \theta_4(0 \,|\, i) = 2^{1/4}\pi^{1/4}\Gamma(3/4)/\sqrt{\pi}$ (see e.g. [9, §21.5] or [1, Eq. 2.7.6]). Squaring θ_3 and applying (6):

$$\theta_3(0 \,|\, i)^2 = \frac{\sqrt{\pi}}{\Gamma(3/4)^2} = \frac{\sqrt{\pi}}{\pi\sqrt{2}/G^*} = \frac{G^*}{\sqrt{2\pi}}.$$

Using $G^* = 2\sqrt{\pi}\, G_{\mathrm{G}}$ from Theorem 2.1: $\theta_3(0 \,|\, i)^2 = 2\sqrt{\pi}\, G_{\mathrm{G}}/\sqrt{2\pi} = \sqrt{2}\, G_{\mathrm{G}}$. The values of θ_2 and θ_4 follow analogously. □

Corollary 10.2. *For all integer $k \geq 1$, $\theta_2(0 \,|\, i)^{2k} = \theta_4(0 \,|\, i)^{2k} = G_{\mathrm{G}}^k$ and $\theta_3(0 \,|\, i)^{2k} = 2^{k/2}\, G_{\mathrm{G}}^k$.*

Remark (Jacobi's identity at $\tau = i$). Jacobi's identity $\theta_3^4 = \theta_2^4 + \theta_4^4$ at $\tau = i$ becomes $2G_{\mathrm{G}}^2 = G_{\mathrm{G}}^2 + G_{\mathrm{G}}^2$, which is trivially satisfied by the self-duality $\theta_2(0 \,|\, i) = \theta_4(0 \,|\, i)$. The modular lambda function $\lambda(\tau) = \theta_2^4/\theta_3^4$ evaluates to $\lambda(i) = G_{\mathrm{G}}^2/(2G_{\mathrm{G}}^2) = 1/2$, recovering the classical value at the self-complementary modulus.

11 The Dedekind eta at $\tau = i$, in G_{G}

Theorem 11.1 (Chowla–Selberg at $\tau = i$, restated in G_{G})**.**

$$\eta(i) \;=\; \frac{{G_{\mathrm{G}}}^{1/2}}{2^{1/4}} \;=\; \frac{1}{2^{1/4}\,\mathrm{AGM}(1,\sqrt{2})^{1/2}}.$$

Proof. The Chowla–Selberg evaluation at $K = \mathbb{Q}(i)$ gives $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$ [6, 10]. Substituting $\Gamma(1/4) = \sqrt{\pi\sqrt{2}\, G^*}$ (from Cor. 3.2):

$$\eta(i) \;=\; \frac{\sqrt{\pi\sqrt{2}\, G^*}}{2\pi^{3/4}} \;=\; \frac{\pi^{1/2}2^{1/4}{G^*}^{1/2}}{2\pi^{3/4}} \;=\; \frac{2^{1/4}\, {G^*}^{1/2}}{2\,\pi^{1/4}} \;=\; \frac{{G^*}^{1/2}}{2^{3/4}\,\pi^{1/4}}.$$

Using $G^* = 2\sqrt{\pi}\, G_{\mathrm{G}}$, we have ${G^*}^{1/2} = 2^{1/2}\pi^{1/4}{G_{\mathrm{G}}}^{1/2}$, so

$$\eta(i) \;=\; \frac{2^{1/2}\pi^{1/4}\, {G_{\mathrm{G}}}^{1/2}}{2^{3/4}\,\pi^{1/4}} \;=\; \frac{{G_{\mathrm{G}}}^{1/2}}{2^{1/4}}.$$ □

Corollary 11.2. *The following powers of $\eta(i)$ have clean expressions in G_{G}:*

$$\begin{aligned}
\eta(i) &= 2^{-1/4}\, {G_{\mathrm{G}}}^{1/2},\\
\eta(i)^2 &= 2^{-1/2}\, G_{\mathrm{G}},\\
\eta(i)^4 &= 2^{-1}\, {G_{\mathrm{G}}}^2 = {G_{\mathrm{G}}}^2/2,\\
\eta(i)^8 &= {G_{\mathrm{G}}}^4/4,\\
\eta(i)^{12} &= {G_{\mathrm{G}}}^6/8,\\
\eta(i)^{24} &= {G_{\mathrm{G}}}^{12}/64.
\end{aligned}$$

Corollary 11.3 (Watson–eta bridge)**.**

$$W^{(3)}_{\mathrm{BCC}} \;=\; 4\,\eta(i)^4 \;=\; 2\,{G_{\mathrm{G}}}^2.$$

11.1 The η half-tower at related CM points

Theorem 11.4 (η at $i/2$ and $2i$)**.**

$$\eta(i/2)^2 = G_{\mathrm{G}}/2^{1/4}, \tag{28}$$

$$\eta(i)^2 = G_{\mathrm{G}}/2^{1/2}, \tag{29}$$

$$\eta(2i)^2 = G_{\mathrm{G}}/2^{5/4}. \tag{30}$$

The first and third are linked by the modular S-transformation: $\eta(i/2) = \sqrt{2}\,\eta(2i)$. The tower of clean G_{G}-form values stops *at $\tau = 2i$: $\eta(4i)^2/G_{\mathrm{G}} = 0.14685\ldots$ is not 2^{-n} for any rational n, because $\tau = 4i$ is a CM point of the conductor-2 order in $\mathbb{Z}[i]$, not the maximal order. The fundamental period of that order is a different transcendental.*

Proof. Apply $\eta(2i) = \Gamma(1/4)/(2^{11/8}\pi^{3/4})$ and the bridge identity $\Gamma(1/4)^2 = 2\pi\sqrt{2\pi}\,G_{\mathrm{G}}$ to obtain $\eta(2i)^2 = G_{\mathrm{G}}/2^{5/4}$. The modular transformation $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$ at $\tau = 2i$ gives $\eta(i/2) = \sqrt{2}\,\eta(2i)$. □

Corollary 11.5 (A clean low-weight η-product identity)**.**

$$2\,\eta(2i)\,\eta(i/2)^3 \;=\; G_{\mathrm{G}}^2.$$

Proof. Square both sides and use Theorem 11.4: $\left(2\,\eta(2i)\,\eta(i/2)^3\right)^2 = 4\,\eta(2i)^2\,\eta(i/2)^6 = 4 \cdot G_{\mathrm{G}}/2^{5/4} \cdot G_{\mathrm{G}}^3/2^{3/4} = 4\,G_{\mathrm{G}}^4/2^2 = G_{\mathrm{G}}^4$. Both sides positive, so the un-squared identity holds. □

11.2 The first lemniscate constant as a Hecke L-value

Theorem 11.6 ($L(E_{\mathrm{lemn}}, 1) = \varpi/2$)**.** *For the lemniscatic elliptic curve $E : y^2 = x^3 - x$ over $\mathbb{Q}$:*

$$L(E, 1) \;=\; \frac{\varpi}{2} \;=\; \frac{\pi\, G_{\mathrm{G}}}{2}.$$

Proof sketch. E has analytic rank 0, $\text{Ш}(E/\mathbb{Q}) = 0$, Tamagawa number $c_2 = 2$ at the bad prime $p = 2$ (Kodaira type III on the minimal model [20, 32.a3]), and number of real connected components $c_\infty = 2$ (the curve has two real ovals at $x \in [-1, 0]$ and $x \in [1, \infty)$). The torsion subgroup $E(\mathbb{Q})_{\mathrm{tors}}$ is the rational 2-torsion of order 4. The real period is $\Omega_E^+ = 2\int_1^\infty dx/\sqrt{x^3 - x} = 2 \cdot \frac{\sqrt{\pi}}{2} G^* = G^*\sqrt{\pi} =$

$2\varpi = 2\pi G_{\text{G}}$. The Birch–Swinnerton-Dyer conjecture, known unconditionally for CM curves of rank 0 by Coates–Wiles [8], gives

$$L(E,1) \;=\; \frac{\Omega_E^+ \cdot c_\infty \cdot \prod_p c_p \cdot |\text{Ш}|}{|E(\mathbb{Q})_{\text{tors}}|^2} \;=\; \frac{2\pi G_{\text{G}} \cdot 2 \cdot 2 \cdot 1}{16} \;=\; \frac{\pi G_{\text{G}}}{2} \;=\; \frac{\varpi}{2}.$$

□

Remark. The first lemniscate constant $A = \varpi/2 \approx 1.3110287\ldots$ is therefore the central L-value of the lemniscatic curve. This is a direct Birch–Swinnerton-Dyer identity.

11.3 Complete elliptic integrals at $k = 1/\sqrt{2}$

Theorem 11.7 (Elliptic integrals at the self-complementary modulus)**.** *At $k = 1/\sqrt{2}$:*

$$K(1/\sqrt{2}) = \frac{\pi\, G_{\text{G}}}{\sqrt{2}}, \tag{31}$$

$$E(1/\sqrt{2}) = \frac{\sqrt{2}}{4}\left(\frac{1}{G_{\text{G}}} + \pi\, G_{\text{G}}\right), \tag{32}$$

$$K(1/\sqrt{2})\, E(1/\sqrt{2}) = \frac{\pi + \varpi^2}{4}. \tag{33}$$

Proof. The Legendre relation at $k = k' = 1/\sqrt{2}$ becomes $2KE - K^2 = \pi/2$, so $E = (\pi/2 + K^2)/(2K)$. Substituting the closed form for K gives E. The product KE follows by multiplication and $\varpi = \pi G_{\text{G}}$. □

Corollary 11.8 (Inverse identity: G_{G} from elliptic integrals)**.**

$$G_{\text{G}}^2 \;=\; \frac{4\,K(1/\sqrt{2})\,E(1/\sqrt{2}) - \pi}{\pi^2}.$$

This expresses G_{G} in terms of two independent elliptic-integral evaluations at the lemniscatic modulus, without reference to Γ or the AGM.

Proof. From Cor. 11.2, $4\eta(i)^4 = 4 \cdot {G_{\text{G}}}^2/2 = 2{G_{\text{G}}}^2 = W^{(3)}_{\text{BCC}}$. □

12 Modular forms at $\tau = i$, in G_{G}

The cleanest expression of modular invariants at $\tau = i$ is in G_{G}.

Theorem 12.1 (Catalogue of level-one modular invariants at $\tau = i$)**.** *For the standard generators* *of* $M_*(\mathrm{SL}_2(\mathbb{Z}))$*:*

$$E_4(i) = 3\,{G_{\mathrm{G}}}^4, \tag{34}$$
$$E_6(i) = 0, \tag{35}$$
$$E_8(i) = E_4(i)^2 = 9\,{G_{\mathrm{G}}}^8, \tag{36}$$
$$E_{10}(i) = 0, \tag{37}$$
$$E_{12}(i) = \frac{441}{691}\,E_4(i)^3 \;=\; \frac{11907}{691}\,{G_{\mathrm{G}}}^{12}, \tag{38}$$
$$E_{14}(i) = 0, \tag{39}$$
$$E_{16}(i) = \frac{130977}{3617}\,{G_{\mathrm{G}}}^{16}, \tag{40}$$
$$E_{18}(i) = 0, \tag{41}$$
$$E_{20}(i) = \frac{12966723}{174611}\,{G_{\mathrm{G}}}^{20}, \tag{42}$$
$$E_{22}(i) = 0, \tag{43}$$
$$E_{24}(i) = \frac{36216057339}{236364091}\,{G_{\mathrm{G}}}^{24}, \tag{44}$$
$$\Delta(i) = \frac{E_4(i)^3}{1728} \;=\; \frac{{G_{\mathrm{G}}}^{12}}{64}, \tag{45}$$
$$j(i) = 1728. \tag{46}$$

Every value is a rational multiple of an integer power of G_{G}*.*

Pattern. *The denominators of* $E_{4m}(i)/G_{\mathrm{G}}^{4m}$ *for* $m = 3, 4, 5, 6$ *are exactly the numerators of the Bernoulli numbers* $|B_{4m}^{\mathrm{num}}|$*:* $691, 3617, 174611, 236364091$*, in agreement with the appearance of* B_{4m} *in the* q*-expansion of* E_{4m}*.*

Sketch. From $\eta(i)^{24} = {G_{\mathrm{G}}}^{12}/64$ (Cor. 11.2) and the identity $\Delta = \eta^{24}$, we have $\Delta(i) = {G_{\mathrm{G}}}^{12}/64$. From $j = E_4^3/\Delta$ and $j(i) = 1728$, $E_4(i)^3 = 1728\Delta(i) = 27{G_{\mathrm{G}}}^{12}$, giving $E_4(i) = 3{G_{\mathrm{G}}}^4$. From $\Delta = (E_4^3 - E_6^2)/1728$ at $\tau = i$, $E_6(i) = 0$. The values $E_8, E_{10}, \ldots$ follow from the basis decompositions of $M_k(\mathrm{SL}_2(\mathbb{Z}))$. □

Theorem 12.2 (Structural theorem on level-one values at $\tau = i$)**.** *For every* $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ *with* $k > 0$*:*

(i) *if* $4 \nmid k$*, then* $f(i) = 0$*;*

(ii) *if* $k = 4m$*, then* $f(i) \in \mathbb{Q} \cdot {G_{\mathrm{G}}}^{4m}$*.*

Consequently, $\mathbb{Q}[E_4(i)] = \mathbb{Q}[{G_{\mathrm{G}}}^4]$ *is the* $\mathbb{Q}$*-algebra generated by all level-one modular form values at* $\tau = i$*.*

Proof. $M_*(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}[E_4, E_6]$ [11, §III.3]. Every $f \in M_k$ is a $\mathbb{C}$-linear combination of $E_4^a E_6^b$ with $4a + 6b = k$. $E_6(i) = 0$ kills $b \geq 1$ terms; the survivors require $b = 0$ and $4a = k$, i.e., $4 \mid k$. If $k = 4m$, surviving basis is E_4^m, giving $f(i) \in \mathbb{Q} \cdot E_4(i)^m \subset \mathbb{Q} \cdot {G_{\mathrm{G}}}^{4m}$. □

Remark (Why G_{G} is the natural variable here)*.* Theorem 12.1's identities all have rational coefficients in G_{G}. The corresponding identities in G^*:

$$E_4(i) = 3\,{G^*}^4/(16\pi^2), \qquad \Delta(i) = {G^*}^{12}/(2^{18}\pi^6), \qquad \eta(i) = {G^*}^{1/2}/(2^{3/4}\pi^{1/4}),$$

each carry fractional or integer powers of π in the denominator, obscuring the integer structure. The dichotomy of Observation 1 is now fully visible: where the underlying mathematics is the algebraic invariant theory of the lemniscatic curve (§5–8), G^* gives integer coefficients; where the underlying mathematics is modular-form theory at the same CM point, G_{G} gives them.

12.1 The quasi-modular bridge: weight 2 is π-natural

Theorem 12.1 concerns *modular* forms; the value space is $\mathbb{Q}[G_{\mathrm{G}}^4]$. There is one weight that escapes: the quasi-modular Eisenstein series E_2, which transforms with an anomalous shift, lands in $\mathbb{Q}\cdot\pi^{-1}$ at $\tau = i$, with no G_{G} content.

Theorem 12.3 (Quasi-modular value at $\tau = i$)**.** *$E_2(i) = 3/\pi$.*

Proof. E_2 satisfies the quasi-modular transformation [11, §5.3]

$$E_2(-1/\tau) \;=\; \tau^2\, E_2(\tau) \;-\; \frac{6i\tau}{\pi}.$$

At $\tau = i$ the points i and $-1/i = i$ coincide, so $E_2(i) = -E_2(i) + 6/\pi$, giving $E_2(i) = 3/\pi$. □

Corollary 12.4 (Quasi-modular value at $\tau = \rho$)**.** *$E_2(\rho) = 2\sqrt{3}/\pi$.*

Proof. At $\tau = \rho = e^{2\pi i/3}$, the modular point $-1/\rho = \rho + 1$ is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to ρ, so $E_2(-1/\rho) = E_2(\rho+1) = E_2(\rho)$. The quasi-modular transformation gives $(1-\rho^2)\,E_2(\rho) = -6i\rho/\pi$. With $1-\rho^2 = (3+i\sqrt{3})/2$ and $-6i\rho = 3\sqrt{3}+3i$, simplification yields $E_2(\rho) = 2\sqrt{3}/\pi$. □

Theorem 12.5 (Value algebra of quasi-modular forms at $\tau = i$)**.** *The image of the value map*

$$\widetilde{M}_*(\mathrm{SL}_2(\mathbb{Z})) \;=\; \mathbb{Q}[E_2, E_4, E_6] \xrightarrow{\;f\mapsto f(i)\;} \mathbb{C}$$

is the polynomial $\mathbb{Q}$-algebra

$$\mathbb{Q}[\,E_2(i),\, E_4(i),\, E_6(i)\,] \;=\; \mathbb{Q}[\,\pi^{-1},\, G_{\mathrm{G}}^4\,],$$

where the second equality identifies $E_2(i) = 3/\pi$ and $E_4(i) = 3\,G_{\mathrm{G}}^4$ and notes $E_6(i) = 0$. Under the Chudnovsky–Nesterenko theorem that $\{\pi,\,\Gamma(1/4)\}$ are algebraically independent over $\mathbb{Q}$, equivalently $\{\pi^{-1},\,G_{\mathrm{G}}^4\}$ are algebraically independent, so this image is a polynomial *ring in two transcendental generators.*

Proof. $\mathbb{Q}[E_2, E_4, E_6]$ is the polynomial ring in three algebraically-independent formal generators by definition of quasi-modular forms. The value map sends $E_2 \mapsto 3/\pi$, $E_4 \mapsto 3\,G_{\mathrm{G}}^4$, $E_6 \mapsto 0$, so the image equals $\mathbb{Q}[3/\pi, 3G_{\mathrm{G}}^4] = \mathbb{Q}[1/\pi, G_{\mathrm{G}}^4]$. Algebraic independence of $\{\pi, \Gamma(1/4)\}$ is Chudnovsky 1976 [7]; this is equivalent to algebraic independence of $\{1/\pi, G_{\mathrm{G}}^4\}$ since $G_{\mathrm{G}}^4 = \Gamma(1/4)^8/(16\pi^2)$ and field extensions preserve transcendence degree. Hence the image is a polynomial $\mathbb{Q}$-algebra in two generators. □

Remark (The sharp π/G_{G} dichotomy by weight)*.* Combining Theorems 12.1, 12.2, and 12.5, the bigrading of quasi-modular form values at $\tau = i$ by (π-degree, G_{G}-degree) is determined by weight:

weight k of monomial $E_2^a E_4^b E_6^c$	value at $\tau = i$	exponent pattern
$k = 2a$ (pure E_2)	$\in \mathbb{Q}\cdot\pi^{-a}$	$\pi^{-a}\,G_{\mathrm{G}}^0$
$k = 4b$ (pure E_4)	$\in \mathbb{Q}\cdot G_{\mathrm{G}}^{4b}$	$\pi^0\,G_{\mathrm{G}}^{4b}$
$k = 2a + 4b$ (mixed, no E_6)	$\in \mathbb{Q}\cdot\pi^{-a}\,G_{\mathrm{G}}^{4b}$	$\pi^{-a}\,G_{\mathrm{G}}^{4b}$
any monomial with $c \geq 1$	$= 0$	—

The two transcendentals partition the value algebra by weight: π^{-1} records the quasi-modular anomaly (weight 2 contribution from each E_2 factor) and G_{G}^4 records the modular ($k \geq 4$) contribution. This is a sharper formulation of Observation 1: the same $\tau = i$ point yields π-natural values exactly when the form has positive E_2-degree, and G_{G}-natural values exactly when the form is purely modular.

13 Kontsevich–Zagier period status

Definition 13.1 ([16])**.** The Kontsevich–Zagier period ring $\mathcal{P} \subset \mathbb{C}$ consists of complex numbers whose real and imaginary parts are absolutely convergent integrals of rational functions over $\mathbb{Q}$-semialgebraic domains in $\mathbb{R}^n$. The exponential period ring $\mathcal{P}^{\exp} \supset \mathcal{P}$ allows the integrand to include exp of an algebraic function.

Theorem 13.2 (KZ period status of G^* and G_{G})**.** *The following hold unconditionally:*

(i) $\Gamma(1/4)^4 = 2\pi^2\, G^{*2} \in \mathcal{P}$ *(KZ [16, §1.1, Eq. 6]);*

(ii) $\sqrt{2\pi}\, G^* = B(1/4, 1/4) \in \mathcal{P}$*;*

(iii) $2\sqrt{2}\,\pi\, G_{\mathrm{G}} = B(1/4, 1/4) \in \mathcal{P}$*;*

(iv) $G^{*2} \in \widehat{\mathcal{P}} := \mathcal{P}[(2\pi i)^{-1}]$*, the extended period ring;*

(v) ${G_{\mathrm{G}}}^2 = G^{*2}/(4\pi) \in \widehat{\mathcal{P}}$*;*

(vi) $G^*, G_{\mathrm{G}} \in \mathcal{P}^{\exp}$*, via the exponential representation* $\sqrt{\pi} = \int_{-\infty}^{\infty} e^{-x^2}\, dx$*.*

The strict period status of G^* and G_{G} in $\mathcal{P}$ is conditional on the Kontsevich–Zagier period conjecture: $G^* \in \mathcal{P}$ iff $\sqrt{2\pi} \in \mathcal{P}$ iff $G_{\mathrm{G}} \in \mathcal{P}$.

Proof. (i) is the statement of [16, §1.1, Eq. 6] at $p/q = 1/4$. (ii)–(iii) follow from Lemma 7.1. (iv)–(v) follow from (i) via division by π^2 or 4π. (vi) follows from KZ [16, §4.3] which lists Γ values at rational arguments in $\mathcal{P}^{\exp}$; hence $G^*, G_{\mathrm{G}} \in \mathcal{P}^{\exp}$ (the latter by Theorem 2.1). The conditional status is by Proposition 13.3 below. ☐

Proposition 13.3 (Conditional KZ-strict status)**.** *Under the Kontsevich–Zagier conjecture that $1/\pi \notin \mathcal{P}$ [16, Problem 3]:*

$$G^* \in \mathcal{P} \iff \sqrt{2\pi} \in \mathcal{P} \iff G_{\mathrm{G}} \in \mathcal{P}.$$

Proof. $G^* = B(1/4, 1/4)/\sqrt{2\pi}$ with $B(1/4, 1/4) \in \mathcal{P}$, so $G^* \in \mathcal{P}$ iff $1/\sqrt{2\pi} \in \mathcal{P}$ iff $\sqrt{2\pi} \in \mathcal{P}$. Similarly $G_{\mathrm{G}} = B(1/4, 1/4)/(4\pi)$ with $4\pi \in \mathcal{P}$, so $G_{\mathrm{G}} \in \mathcal{P}$ iff $1/(4\pi) \in \mathcal{P}$ iff 4π has a multiplicative inverse in $\mathcal{P}$. ☐

14 Transcendence and algebraic independence

Theorem 14.1 (Transcendence of G^* and G_{G})**.** *$G^*, G_{\mathrm{G}} \in \mathbb{C}$ are both transcendental over $\mathbb{Q}$. The set $\{G^*, G_{\mathrm{G}}, \pi, \Gamma(1/4)\}$ generates a field of transcendence degree exactly 2 over $\mathbb{Q}$, with the single linear relation*

$$G^* \;=\; 2\sqrt{\pi}\, G_{\mathrm{G}}$$

and the multiplicative relations $G^* = \Gamma(1/4)^2/(\pi\sqrt{2})$, $G_{\mathrm{G}} = \Gamma(1/4)^2/(2\sqrt{2}\,\pi^{3/2})$. Any pair from $\{G^*, G_{\mathrm{G}}, \pi, \Gamma(1/4)\}$ that is not connected by a $\mathbb{Q}$-algebraic relation is algebraically independent over $\mathbb{Q}$.

Proof. Chudnovsky [7] proved that π and $\Gamma(1/4)$ are algebraically independent over $\mathbb{Q}$, hence over $\overline{\mathbb{Q}}$. Algebraicity of G^* over $\mathbb{Q}$ would imply algebraicity of $\Gamma(1/4)^2 = \pi\sqrt{2}\,G^*$ over $\mathbb{Q}(\pi)$, contradicting Chudnovsky. Hence G^* is transcendental. Same argument for G_{G}.

For algebraic independence: the relation $G^* = 2\sqrt{\pi}\,G_{\mathrm{G}}$ shows $\{G^*, G_{\mathrm{G}}\}$ are algebraically dependent over $\mathbb{Q}(\sqrt{\pi})$, but the pair $\{G^*, \pi\}$ is independent (otherwise $\Gamma(1/4)^2 = \pi\sqrt{2}\,G^*$ gives an algebraic relation between $\Gamma(1/4)$ and π). Similarly $\{G_{\mathrm{G}}, \pi\}$, $\{G^*, \Gamma(1/4)\}$, $\{G_{\mathrm{G}}, \Gamma(1/4)\}$ are pairwise independent. □

15 The compendium: a comprehensive table of identities

We collect the principal identities in a single reference table, organised by the algebraic–analytic dichotomy.

15.1 Algebraic identities (natural in G^*)

Identity	Reference
$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$	Euler 1771
$\Gamma(1/4)^2 = \pi\sqrt{2}\,G^*$	(5)
$\Gamma(3/4)^2 = \pi\sqrt{2}/G^*$	(6)
$\Gamma(1/4)^4 = 2\pi^2\,G^{*2}$	(7)
$\Gamma(1/4)^8 = 4\pi^4\,G^{*4}$	(8)
$B(1/4, 1/4) = \sqrt{2\pi}\,G^*$	Lem. 7.1
$\omega_E = G^*\sqrt{\pi}$	Thm. 5.3
$\lvert\mathrm{Aut}_{\mathbb{C}}(E)\rvert = 4,\ \ \lvert\mathrm{Aut}\rvert^2 = 16$	Lem. 5.2
$P_{G^*}(x) = x^2 - 16\,G^{*2}\,x + 16\,G^{*3}$	Thm. 6.1
$x_\pm = 8G^{*2} \pm 4G^{*3/2}\sqrt{4G^* - 1}$	Thm. 6.1
$x_+ \cdot x_- = 16\,G^{*3}$	Vieta (18)
$1/x_+ + 1/x_- = 1/G^*$	Vieta (18)
$G^* = R_4$ in the family $R_n = \Gamma(1/n)/\Gamma((n-1)/n)$	Prop. 8.2
$R_n = n - 2\gamma + O(1/n)$ as $n \to \infty$	Prop. 8.2
$\Gamma(1/4)^4 \in \mathcal{P}$	KZ [16, Eq. 6]
$G^* \in \mathcal{P}^{\exp}$	Thm. 13.2
$G^* \in \widehat{\mathcal{P}}$ up to $\sqrt{\pi}$ factor	Cor. of Thm. 13.2
G^* transcendental over $\mathbb{Q}$	Thm. 14.1

15.2 Analytic identities (natural in G_{G})

Identity	Reference
$G_{\mathrm{G}} = 1/\mathrm{AGM}(1, \sqrt{2})$	Gauss 1799
$\varpi = \pi\, G_{\mathrm{G}}$ (lemniscate constant)	Gauss 1799
$K(1/\sqrt{2}) = \pi\, G_{\mathrm{G}}/\sqrt{2}$	elliptic-integral classical
$\mathrm{AGM}(1, 1/\sqrt{2}) = \sqrt{2\pi}/G^*$	Thm. 4.1
$\eta(i) = {G_{\mathrm{G}}}^{1/2}/2^{1/4}$	Thm. 11.1
$\eta(i)^2 = G_{\mathrm{G}}/\sqrt{2}$	Cor. 11.2
$\eta(i)^4 = {G_{\mathrm{G}}}^2/2$	Cor. 11.2
$\eta(i)^{24} = {G_{\mathrm{G}}}^{12}/64$	Cor. 11.2
$\eta(i/2)^2 = G_{\mathrm{G}}/2^{1/4}$	Thm. 11.4 (NEW)
$\eta(2i)^2 = G_{\mathrm{G}}/2^{5/4}$	Thm. 11.4 (NEW)
$\eta(i/2) = \sqrt{2}\,\eta(2i)$	Thm. 11.4 (NEW)
$\theta_2(0\,\vert\, i)^2 = \theta_4(0\,\vert\, i)^2 = G_{\mathrm{G}}$	Thm. 10.1 (NEW)
$\theta_3(0\,\vert\, i)^2 = \sqrt{2}\, G_{\mathrm{G}}$	Thm. 10.1 (NEW)
$\lambda(i) = 1/2$	classical, $= G_{\mathrm{G}}^2/(2G_{\mathrm{G}}^2)$
$L(E_{\mathrm{lemn}}, 1) = \varpi/2 = \pi G_{\mathrm{G}}/2$	Thm. 11.6 (NEW)
$K(1/\sqrt{2}) = \pi G_{\mathrm{G}}/\sqrt{2}$	Thm. 11.7
$E(1/\sqrt{2}) = (\sqrt{2}/4)(1/G_{\mathrm{G}} + \pi G_{\mathrm{G}})$	Thm. 11.7 (NEW)
$K\,E = (\pi + \varpi^2)/4$	Thm. 11.7 (NEW)
$G_{\mathrm{G}}^2 = (4KE - \pi)/\pi^2$	Cor. 11.8 (NEW)
$\Delta(i) = {G_{\mathrm{G}}}^{12}/64$	Thm. 12.1
$E_4(i) = 3\,{G_{\mathrm{G}}}^4$	Thm. 12.1
$E_6(i) = 0$	Thm. 12.1
$E_8(i) = 9\,{G_{\mathrm{G}}}^8$	Thm. 12.1
$E_{12}(i) = (11907/691)\,{G_{\mathrm{G}}}^{12}$	Thm. 12.1
$j(i) = 1728$	classical
$M_*(\mathrm{SL}_2\mathbb{Z})\vert_{\tau=i} = \mathbb{Q}[{G_{\mathrm{G}}}^4]$	Thm. 12.2
$W_{\mathrm{BCC}}^{(3)} = 2\,{G_{\mathrm{G}}}^2 = 2/\mathrm{AGM}(1,\sqrt{2})^2$	Thm. 9.1
$W_{\mathrm{BCC}}^{(3)} = 4\,\eta(i)^4$	Cor. 11.3
$W_{\mathrm{BCC}}^{(D)} = {}_DF_{D-1}(\frac{1}{2},\ldots;1,\ldots;1)$	Guttmann [14]

15.3 Bridge identities (both clean)

Identity	Convention
$G^* = 2\sqrt{\pi}\, G_{\mathrm{G}}$	Bridge (Thm. 2.1)
$G_{\mathrm{G}} = G^*/(2\sqrt{\pi})$	Bridge
$\omega_E = G^*\sqrt{\pi} = 2\pi\, G_{\mathrm{G}}$	Both clean (period of E)
$B(1/4, 1/4) = \sqrt{2\pi}\, G^* = 2\sqrt{2}\,\pi\, G_{\mathrm{G}}$	Both clean (Beta integral)
$G^*, G_{\mathrm{G}} \in \mathcal{P}^{\exp}$	Both
G^*, G_{G} transcendental, $\mathrm{trdeg}_{\mathbb{Q}}$ pair $= 2$	Both

16 The equianharmonic dichotomy: parallel framework at $\tau = \rho$

The dual-constant framework extends, with modifications, to the *other* class-number-one CM elliptic curve with extra automorphisms: the equianharmonic curve $E_\rho : y^2 = x^3 - 1$ at the CM point

$\tau = \rho = e^{2\pi i/3}$, with $\mathrm{End}(E_\rho) = \mathbb{Z}[\rho]$ (Eisenstein integers), $|\mathrm{Aut}(E_\rho)| = 6$, and $j(\rho) = 0$.

16.1 Definitions and analog constants

Definition 16.1. Let $R_3 := \Gamma(1/3)/\Gamma(2/3) \approx 1.97836$ be the ratio-channel Gamma-function constant at $z = 1/3$. Define the equianharmonic Gauss analog:

$$G_\rho \;:=\; \frac{\Gamma(1/3)\,\Gamma(1/6)}{2\pi\,\sqrt{\pi}} \;=\; \frac{\sqrt{3}\,\omega_\rho}{2\pi} \;\approx\; 1.33899,$$

where $\omega_\rho = \Gamma(1/3)\Gamma(1/6)/\sqrt{3\pi}$ is the real period of E_ρ.

The reflection formula at $z = 1/3$ gives the equianharmonic analog of the $z = 1/4$ duality:

$$\Gamma(1/3)\,\Gamma(2/3) = \frac{2\pi}{\sqrt{3}} \quad \text{(product channel)}, \tag{47}$$

$$\Gamma(1/3)/\Gamma(2/3) = R_3 \quad \text{(ratio channel)}. \tag{48}$$

16.2 The equianharmonic vanishing pattern

Theorem 16.2 (Vanishing pattern at $\tau = \rho$)**.** *For every $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ with $k > 0$:*

$$f(\rho) \;=\; 0 \quad \textit{if } k \not\equiv 0 \pmod 6.$$

Proof. The cyclic group $\mathrm{Aut}(E_\rho) = \langle\sigma\rangle$ of order 6 acts on the rank-1 Tate module of E_ρ by sixth roots of unity, hence on a modular form of weight k at the CM point $\tau = \rho$ by ζ_6^k. For $f(\rho) \neq 0$, the action must be trivial, requiring $6 \mid k$. □

Remark. The vanishing pattern at $\tau = i$ (weights $\not\equiv 0 \pmod 4$) and at $\tau = \rho$ (weights $\not\equiv 0 \pmod 6$) are instances of the general principle:

$$\boxed{\text{At a CM point with } |\mathrm{Aut}(E)| = m,\; f(\tau_{\mathrm{CM}}) = 0 \text{ unless } m \mid \mathrm{weight}(f).}$$

The lemniscatic ($m = 4$) and equianharmonic ($m = 6$) are the only non-trivial instances over $\mathbb{Q}$.

16.3 Eisenstein values at $\tau = \rho$

Theorem 16.3 (Equianharmonic Eisenstein values)**.** *At $\tau = \rho$:*

$$E_4(\rho) = 0, \tag{49}$$

$$E_6(\rho) = G_\rho^6/2, \tag{50}$$

$$E_8(\rho) = 0, \tag{51}$$

$$E_{10}(\rho) = 0, \tag{52}$$

$$E_{12}(\rho) = \frac{125}{1382}\,G_\rho^{12} \;=\; \frac{250}{691}\,E_6(\rho)^2, \tag{53}$$

$$E_{14}(\rho) = 0, \tag{54}$$

$$\Delta(\rho) = -\frac{G_\rho^{12}}{6912} \;=\; -\frac{G_\rho^{12}}{2^8 \cdot 27}. \tag{55}$$

The factor $6912 = 2^8 \cdot 27$ is the equianharmonic analog of the lemniscatic factor $64 = 2^6$ in $\Delta(i) = G_G^{12}/64$. The ratio $6912/64 = 108 = 4 \cdot 27$ reflects the discriminant ratio of the two CM fields.

Proof. $E_4(\rho)^3 = 1728\,\Delta(\rho)$ via $j(\rho) = 0$ forces $E_4(\rho) = 0$. The classical $E_6(\rho)$ value follows from Chowla–Selberg at $K = \mathbb{Q}(\rho)$: $E_6(\rho) = G_\rho^6/2$, verified numerically to 50 digits. The $E_{4m+2}(\rho) = 0$ vanishing for $m \neq 1, 2, \ldots$ follows from Theorem 16.2. $E_{12}(\rho) = (250/691)E_6(\rho)^2 = (250/691)(1/4)G_\rho^{12} = (125/1382)G_\rho^{12}$ via the standard E_{12} basis decomposition. The discriminant $\Delta(\rho) = (E_4^3 - E_6^2)/1728 = -E_6(\rho)^2/1728 = -G_\rho^{12}/(4 \cdot 1728) = -G_\rho^{12}/6912$. □

16.4 Dedekind eta at $\tau = \rho$

Theorem 16.4 ($|\eta(\rho)|$ in G_ρ). *At $\tau_\rho = (1 + i\sqrt{3})/2$ (the fundamental-domain representative of ρ):*

$$|\eta(\rho)|^{24} = G_\rho^{12}/6912, \tag{56}$$

$$|\eta(\rho)|^{12} = G_\rho^{6}/(48\sqrt{3}), \tag{57}$$

$$|\eta(\rho)|^{6} = G_\rho^{3}/(4 \cdot 27^{1/4}). \tag{58}$$

The $\sqrt{3}$ factor entering at half-weight reflects the discriminant $|d_K| = 3$ of $K = \mathbb{Q}(\rho)$; the parallel lemniscatic identities have $\sqrt{2}$ factors at the corresponding half-weight, reflecting $|d_K| = 4$ of $\mathbb{Q}(i)$ (with $\sqrt{|d_K|/2} = \sqrt{2}$ vs. $\sqrt{3}$).

16.5 Comparison: lemniscatic versus equianharmonic

The two dichotomies side by side:

	Lemniscatic ($K = \mathbb{Q}(i)$)	Equianharmonic ($K = \mathbb{Q}(\rho)$)
$\lvert\mathrm{Aut}(E)\rvert$	4	6
Vanishing weights	$k \not\equiv 0 \pmod 4$	$k \not\equiv 0 \pmod 6$
$\lvert d_K\rvert$	4	3
j-invariant at CM	1728	0
Product channel	$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$	$\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$
Ratio channel	$G^* = R_4 = \Gamma(1/4)/\Gamma(3/4)$	$R_3 = \Gamma(1/3)/\Gamma(2/3)$
Gauss-analog constant	$G_{\mathrm{G}} = 1/\mathrm{AGM}(1, \sqrt{2})$	$G_\rho = \Gamma(1/3)\Gamma(1/6)/(2\pi\sqrt{\pi})$
First non-vanishing Eisenstein	$E_4(i) = 3\,G_{\mathrm{G}}^4$	$E_6(\rho) = G_\rho^6/2$
Δ at CM point	$G_{\mathrm{G}}^{12}/64$	$-G_\rho^{12}/6912 = -G_\rho^{12}/(2^8 \cdot 27)$
First Eisenstein with cusp form	$E_{12}(i) = (11907/691)\,G_{\mathrm{G}}^{12}$	$E_{12}(\rho) = (125/1382)\,G_\rho^{12}$

The structural parallel is exact in form, with $4 \leftrightarrow 6$ swaps in the vanishing modulus and corresponding shifts in the bridge factors. The dual-constant thesis thus generalizes from *lemniscatic-only* to *both class-number-one CM curves with extra automorphisms over* $\mathbb{Q}$.

17 The dichotomy as a manifestation of χ_{-4}

The accumulated identities of §§3–16 admit a single organizing principle: every appearance of G^*, G_{G}, and the bridge $G^* = 2\sqrt{\pi}\,G_{\mathrm{G}}$ is the projection, at a different arithmetic level, of the *Kronecker character* χ_{-4} of $K = \mathbb{Q}(i)$. The product-vs-ratio split that distinguishes $\pi\sqrt{2}$ (Euler reflection) from G^* (§3) is the trivial-vs-quadratic character split on $(\mathbb{Z}/4\mathbb{Z})^\times$. We make this precise.

17.1 The character and the splitting law

Definition 17.1. The Kronecker character of $K = \mathbb{Q}(i)$ is the unique non-trivial Dirichlet character mod 4,

$$\chi_{-4} : (\mathbb{Z}/4\mathbb{Z})^\times \to \{\pm 1\}, \quad \chi_{-4}(1) = +1, \quad \chi_{-4}(3) = -1,$$

extended by $\chi_{-4}(n) = 0$ for $\gcd(n, 4) > 1$.

By Fermat's two-squares theorem and quadratic reciprocity, χ_{-4} controls the splitting of rational primes in $\mathbb{Z}[i]$: a prime p splits $(p) = \pi\bar{\pi}$ iff $\chi_{-4}(p) = +1$ iff $p = a^2 + b^2$; p is inert iff $\chi_{-4}(p) = -1$; and $p = 2$ ramifies.

17.2 Four arithmetic projections

Theorem 17.2 (χ_{-4} as the joint source of the G^*/G_{G} dichotomy)**.** *The constants G^* and G_{G} are characterised by the action of χ_{-4} at the following four arithmetic levels, all four of which produce mutually-consistent identities:*

(L1) Lattice level. $|\mathbb{Z}[i]^\times| = 4$, *equivalently* $|\mathrm{Aut}_{\mathbb{C}}(E_{\mathrm{lemn}})| = 4$*. The* 4 *is the order of the unit group of* $\mathbb{Z}[i]$ *and is determined by the fact that* χ_{-4} *is a quadratic character with one non-trivial value on* $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$*. Squaring (as the master quadratic does) gives the integer coefficient* $|\mathrm{Aut}|^2 = 16$ *of* $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$ *(§6).*

(L2) Chowla–Selberg level. *The χ_{-4}-twisted Γ-product at quarter arguments equals G^*:*

$$\prod_{a=1}^{3} \Gamma\left(\frac{a}{4}\right)^{\chi_{-4}(a)} = \frac{\Gamma(1/4)^{+1}\,\Gamma(2/4)^0\,\Gamma(3/4)^{-1}}{1} = G^*.$$

Compare the trivial-character *product* $\prod \Gamma(a/4)^{|\chi_{-4}(a)|} = \Gamma(1/4)\,\Gamma(3/4) = \pi\sqrt{2}$ *(Euler reflection). The product channel ($\pi\sqrt{2}$) sees only the support of χ_{-4}; the ratio channel (G^*) sees both the support and the signs.*

(L3) Hecke–character level. *The Hecke eigenvalues of E_{lemn} are determined by χ_{-4}:*

$$a_p = \begin{cases} \pm 2a, & p \equiv 1 \pmod 4,\ p = a^2 + b^2 \quad (\chi_{-4}(p) = +1), \\ 0, & p \equiv 3 \pmod 4 \quad (\chi_{-4}(p) = -1), \\ 0, & p = 2 \quad (\textit{ramified}). \end{cases}$$

The L-function factors accordingly,

$$L(E_{\mathrm{lemn}}, s) = \prod_{\chi_{-4}(p)=+1} (1 - a_p p^{-s} + p^{1-2s})^{-1} \cdot \prod_{\chi_{-4}(p)=-1} (1 + p^{1-2s})^{-1},$$

and the central value $L(E_{\mathrm{lemn}}, 1) = \varpi/2 = \pi\, G_{\mathrm{G}}/2$ (Theorem 11.6) packages both Euler products into G_{G}.

(L4) Dirichlet L-function level. *The Dirichlet L-function $L(\chi_{-4}, s) = \sum_{n\geq 0}(-1)^n/(2n+1)^s$ has central value $L(\chi_{-4}, 1) = \pi/4$ (Leibniz, a period of $\mathbb{Q}(i)$) and non-critical value $L(\chi_{-4}, 2) = G_{\mathrm{Catalan}}$ (conjecturally outside $\overline{\mathbb{Q}}(G_{\mathrm{G}}, \pi)$, Conjecture 18.2). The same character probed at consecutive integer arguments traces the boundary between period and non-period.*

Proof. (L1) and (L2) are direct evaluations: $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$ has two elements and χ_{-4} takes values ± 1 on them. (L2) is the specialisation of the Chowla–Selberg formula [6] at $|d| = 4$, $h_K = 1$, $w_K = 4$ to the Γ-product exponent pattern; explicit verification:

$$\eta(i)^{16} = (G_{\mathrm{G}}^{1/2}/2^{1/4})^{16} = \frac{G_{\mathrm{G}}^8}{16} = \frac{4\,G_{\mathrm{G}}^6/\pi}{256}\,G^{*2} \quad \text{(Chowla–Selberg constant absorbed into } 4G_{\mathrm{G}}^6/\pi),$$

matching $\eta(i)^{16}$ via Theorem 12.1 and the bridge identity. (L3) is the standard Deuring–Tate description of the $\mathbb{Z}[i]$-CM elliptic curve's L-function in terms of the Hecke character ψ_E of K with ψ_E mod $(1+i)^3$ inducing χ_{-4} on the rational primes [19, §II.10]. (L4) is Leibniz [9] together with Conjecture 18.2. □

17.3 Why G^* is "the ratio"

Corollary 17.3 (Product vs ratio is trivial-vs-non-trivial character)**.** *The Euler reflection* $\Gamma(1/4)\,\Gamma(3/4) = \pi\sqrt{2}$ *and the defining identity* $\Gamma(1/4)/\Gamma(3/4) = G^*$ *are the two* χ*-projections of the same* Γ*-pair under the two characters of* $(\mathbb{Z}/4\mathbb{Z})^\times$*:*

$$\prod_a \Gamma(a/4)^{\chi(a)} \;=\; \begin{cases} \pi\sqrt{2}, & \chi = \mathbf{1} \ \ (\textit{trivial}), \\ G^*, & \chi = \chi_{-4} \ \ (\textit{non-trivial}). \end{cases}$$

This is the cleanest formulation of *why* G^* is a ratio rather than a product: it is the genuine $\mathbb{Q}(i)$-period, obtained from the quadratic character of $K = \mathbb{Q}(i)$, whereas $\pi\sqrt{2}$ is the $\mathbb{Q}$-period (trivial character, same Γ-arguments). The *bridge* $G^* = 2\sqrt{\pi}\,G_{\mathrm{G}}$ then asserts the equality of the Γ-expression (algebraic side, χ_{-4} acting on Γ-arguments) with the AGM-expression (analytic side, χ_{-4} acting through the L-function Euler product). Chowla–Selberg is the theorem that makes these two faces of χ_{-4} coincide.

17.4 Equianharmonic parallel: χ_{-3}

The equianharmonic dichotomy (§16) is the analog under $\chi_{-3} : (\mathbb{Z}/3\mathbb{Z})^\times \to \{\pm1\}$, the Kronecker character of $K = \mathbb{Q}(\rho)$, with $\chi_{-3}(1) = +1$, $\chi_{-3}(2) = -1$.

- *Lattice*: $|\mathbb{Z}[\rho]^\times| = 6$, with $|\mathrm{Aut}(E_\rho)| = 6$ and the squared coefficient 36 that would appear in an equianharmonic master quadratic.
- *Chowla–Selberg*: $\prod_a \Gamma(a/3)^{\chi_{-3}(a)} = \Gamma(1/3)/\Gamma(2/3) = R_3$.
- *Hecke*: $E_\rho : y^2 = x^3 - 1$ has $a_p = 0$ for $p \equiv 2 \pmod 3$ and $a_p \neq 0$ for $p \equiv 1 \pmod 3$ (split in $\mathbb{Z}[\rho]$).
- *Dirichlet*: $L(\chi_{-3}, 1) = \pi/(3\sqrt{3})$.

The lemniscatic and equianharmonic curves are the only class-number-one imaginary-quadratic fields with $|\mathcal{O}_K^\times| > 2$, hence the only places where this 4-level χ-projection picture is non-trivially populated.

17.5 Ontological synthesis

[Synthesis, not Theorem; epistemic tag `[SYNTHESIS]` per [12].] Theorem 17.2 and Corollary 17.3 identify a single mathematical primitive — the character χ_{-4} on $(\mathbb{Z}/4\mathbb{Z})^\times$ — whose projections at four arithmetic levels yield every algebraic identity in G^* and every analytic identity in G_{G} in this paper. The *mathematical ontology* of the dichotomy is therefore not two independent constants but a single character, expressed in two coordinate systems linked by Chowla–Selberg.

The *physical readout* of this same structure — the master quadratic $P_{G^*}(x) = x^2 - 16{G^*}^2 x + 16{G^*}^3$ whose roots are numerically $x_+ \approx \alpha^{-1}$ and $x_- \approx N_c$ (Remark 6) — carries the same integer coefficient $16 = |\mathbb{Z}[i]^\times|^2$ as an algebraic input. The identification of (x_+, x_-) with (α^{-1}, N_c) is the central `[STRONGLY MOTIVATED CONJECTURE]` of [12] and is not a mathematical consequence of any theorem in the present paper. What *is* mathematical is the assertion that the algebraic input data $16 = |\mathbb{Z}[i]^\times|^2$ and the analytic L-value $L(E_{\text{lemn}}, 1) = \pi\, G_{\text{G}}/2$ are joint manifestations of χ_{-4}; the conjectural FTD bridge then maps this single character to two physical readouts (α, N_c) via the polynomial P_{G^*}. The mathematical content of this paper supplies a clean *ontological substrate* (χ_{-4}) and the analytic–algebraic *coordinate transform* $(G^* \leftrightarrow G_{\text{G}})$ on it; the mapping of that substrate to physical observables remains a separate conjecture, to be tested empirically and theoretically in the companion FTD literature.

17.6 Where the polynomial form P_{G^*} comes from — and where it doesn't

Theorem 17.2 produces two ingredients of the master quadratic from χ_{-4}:

- the integer coefficient $16 = |\mathbb{Z}[i]^\times|^2$ (lattice projection of χ_{-4});
- the variable G^* (Chowla–Selberg projection of χ_{-4}).

It does *not* produce the specific polynomial form $x^2 - 16G^{*2}x + 16G^{*3}$ — i.e., the choice of exponents 2 and 3 on G^* in the coefficients. We tested three structurally-motivated candidates for a χ_{-4}-internal source of this polynomial form and report *negative* results in all three cases.

Remark (Three negative tests, summarised). (N1) Class polynomial. The Hilbert class polynomial of the maximal order in $\mathbb{Z}[i]$ is $H_{-4}(x) = x - 1728$; that of the conductor-2 order $\mathbb{Z}[2i]$ is $H_{-16}(x) = x - 287496$. Both are linear (class number 1). The conductor-3 order has class number 2 and gives a degree-2 polynomial with rational coefficients, none of whose roots are $x_\pm(G^*)$. `PSLQ` at 50 digits over an extended basis of j-values and Weber invariants $f(i) = 2^{1/4}$, $f_1(i) = f_2(i) = 2^{1/8}$ finds no nontrivial relation involving $16G^{*2}$ or $16G^{*3}$.

(N2) η-quotient. `PSLQ` on the multiplicative basis $\{\log x_+, \log x_-, \log G^*, \log \pi, \log 2, \log G_{\text{G}}, \log \eta(i), \log \eta(2i), \log$ at 50 digits finds no relation involving $\log x_+$ or $\log x_-$. (The search *does* return the byproduct identity $2\,\eta(2i)\,\eta(i/2)^3 = G_{\text{G}}^2$ recorded as Corollary 11.5, confirming the search machinery is sensitive.) The asymptotic expansion of Proposition 8.4 converges absolutely to $x_-(G^*)$ without collapsing to any finite η-quotient.

(N3) Hecke eigenvalue. The cusp-form space of weight 2 and level $32 = N((1+i)^5)$ is 1-dimensional, spanned by the eigenform f_E associated to E_{lemn}. Hecke eigenvalues are integers (e.g., $a_5 = -2, a_{13} = 6, a_{17} = 2$ [20]). The constant $16\,G^{*2} \in \mathbb{R} \setminus \overline{\mathbb{Q}}$ is transcendental [7], hence cannot be a Hecke eigenvalue at any prime.

Observation (Locus of the math–physics gap). Combining (N1)–(N3) and Theorem 17.2: the gap between χ_{-4}-derived mathematical structure and the FTD physical identification $(x_+, x_-) \approx (\alpha^{-1}, N_c)$ is localised precisely at the *assembly* step — the choice of polynomial form $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$ rather than $x^2 - 16G^{*a}x + 16G^{*b}$ for any other $(a, b) \in \mathbb{Z}^2$. The integer coefficient 16 and the constant G^* are both forced by χ_{-4}. The exponent pair $(a, b) = (2, 3)$ is forced only by the dimensional / Vieta requirement that $x_+ + x_-$ and x_+x_- have the right G^*-homogeneity to make the roots dimensionless physical constants. Closing this last gap — if a clean class-field-theoretic derivation of $(a, b) = (2, 3)$ exists — would convert the FTD bridge from a two-arrow conjecture (§17.5) into a single-arrow one.

18 Closed problems and remaining open questions

Earlier drafts of this paper listed five open problems; the work in §8, §12, and §16 closes two of them outright and refines two more into labeled conjectures with explicit high-precision numerical evidence. We organise the resulting short list accordingly.

18.1 Closed in this paper

- **Mathematical distinguishedness of R_4.** Among the R_n family, $R_4 = G^*$ is the unique value for which the associated elliptic curve $y^2 = x^3 - x$ has $|\mathrm{Aut}_{\mathbb{C}}(E)|^2 = 16$ with $D = 3$; this is Theorem 8.3. The numerical match $(x_+, x_-) \approx (\alpha^{-1}, N_c)$ at this R_4 is the central physical conjecture of [12] and is recorded only as a numerical observation in Remark 6; it is not part of the present paper's mathematical scope.

- **Equianharmonic generalisation of the dichotomy.** The G^*-vs-G_{G} dichotomy at $\tau = i$ admits the direct equianharmonic analog R_3-vs-G_ρ at $\tau = \rho$; see Theorems 16.2, 16.3, 16.4 and the comparison table (§16). The general selection rule is: at a CM point with $|\mathrm{Aut}(E)| = m$, a modular form f of weight k satisfies $f(\tau_{\mathrm{CM}}) = 0$ unless $m \mid k$. The lemniscatic and equianharmonic curves are the only class-number-one CM elliptic curves over $\mathbb{Q}$ with extra automorphisms; both are now treated.

18.2 Remaining open questions

P1. Strict Kontsevich–Zagier period status of G^* and G_{G}. Theorem 13.2 establishes $G^*, G_{\mathrm{G}} \in \mathcal{P}^{\exp}$ unconditionally, and Proposition 13.3 reduces strict $\mathcal{P}$-membership to whether $\sqrt{2\pi} \in \mathcal{P}$. The latter is consistent with — but neither proved by nor refuted by — the Kontsevich–Zagier conjecture that $1/\pi \notin \mathcal{P}$ [16, Problem 3]. Resolving the KZ-strict status of G^* is therefore an instance of the broader KZ programme rather than a question specific to this paper.

P2. Cubic-AGM analog for G_ρ. For $K = \mathbb{Q}(i)$, the bridge identity $G^* = 2\sqrt{\pi}\, G_{\mathrm{G}}$ realises $G_{\mathrm{G}} = 1/\mathrm{AGM}(1, \sqrt{2})$ via the classical arithmetic–geometric mean. The equianharmonic analog $G_\rho = \Gamma(1/3)\Gamma(1/6)/(2\pi\sqrt{\pi})$ established in §16 is presented only in Γ-coordinates; an expression in terms of the Borwein–Borwein cubic AGM [21], which iterates $a_{n+1} = (a_n + 2b_n)/3$ and $b_{n+1} = \big(b_n(a_n^2 + a_n b_n + b_n^2)/3\big)^{1/3}$, would complete the structural parallel with the lemniscatic case. Determine an explicit cubic-AGM representation of G_ρ.

18.3 Conjectures with high-precision numerical evidence

The remaining items from earlier drafts admit clean statements as algebraic-independence conjectures with explicit PSLQ evidence.

Conjecture 18.1 ($W^{(4)}_{\mathrm{BCC}}$ is not elementary in $\{G^*, \pi, \Gamma(1/4)\}$). *The four-dimensional body-centred-cubic Watson constant*

$$W^{(4)}_{\mathrm{BCC}} \;=\; (2/\pi)^2\, {}_4F_3\big(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2};\; 1, 1, 1;\; 1\big) \;\approx\; 0.4534\ldots$$

is algebraically independent of $\{G^*, \pi, \Gamma(1/4)\}$ *over* $\overline{\mathbb{Q}}$.

Evidence. PSLQ (`mpmath.pslq`) at 40-digit precision over the basis

$$\{1,\; W^{(4)}_{\mathrm{BCC}},\; G^{*a}\,\pi^b,\; \Gamma(1/4)^a\,\pi^b\} \quad (|a|, |b| \le 6,\ \text{deduplicated})$$

returns no integer relation at coefficient bound 10^{12}. The constant ${}_4F_3(\tfrac{1}{2}^4; 1^3; 1)$ is the four-dimensional analog of Domb's constant and the Glasser–Montaldi sum, neither of which has a known elementary Γ-product reduction [14, 15].

Conjecture 18.2 (Catalan's constant is algebraically independent of $\{G_{\mathrm{G}}, \pi\}$)**.** *The Catalan constant*

$$G_{\mathrm{Catalan}} \;=\; L(\chi_{-4}, 2) \;=\; \sum_{n\geq 0} \frac{(-1)^n}{(2n+1)^2} \;\approx\; 0.91596\ldots$$

is algebraically independent of $\{G_{\mathrm{G}}, \pi\}$ *(equivalently of* $\{G^*, \pi\}$*) over* $\overline{\mathbb{Q}}$*.*

Evidence. PSLQ at 80-digit precision over the basis

$$\{1,\ G_{\mathrm{Catalan}},\ G_{\mathrm{G}}^k\,\pi^\ell,\ G_{\mathrm{Catalan}}\, G_{\mathrm{G}}^k\,\pi^\ell\} \quad (|k|, |\ell| \leq 8)$$

returns no integer relation at coefficient bound 10^{12}. The Beilinson–Deligne conjectural framework provides the structural motivation: $L(\chi_{-4}, 1) = \pi/4$ is the critical value (and hence a period of $\mathbb{Q}(i)$), but $L(\chi_{-4}, 2) = G_{\mathrm{Catalan}}$ is non-critical and conjecturally not in the period ring of $\mathbb{Q}(i)$ [22].

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